

Probability and Probability distribution

Introduction to Probability

Descriptive Statistics: Numerical Measures

- ❑ Experiments, Counting Rules, and Assigning Probabilities
- ❑ Events and Their Probability
- ❑ Relationships of Probability
- ❑ Conditional Probability
- ❑ Bayes' Theorem

Uncertainties

Managers often base their decisions on an analysis of uncertainties such as the following:

What are the chances that sales will decrease if we increase prices?

What is the likelihood a new assembly method will increase productivity?

What are the odds that a new investment will be profitable?

Probability

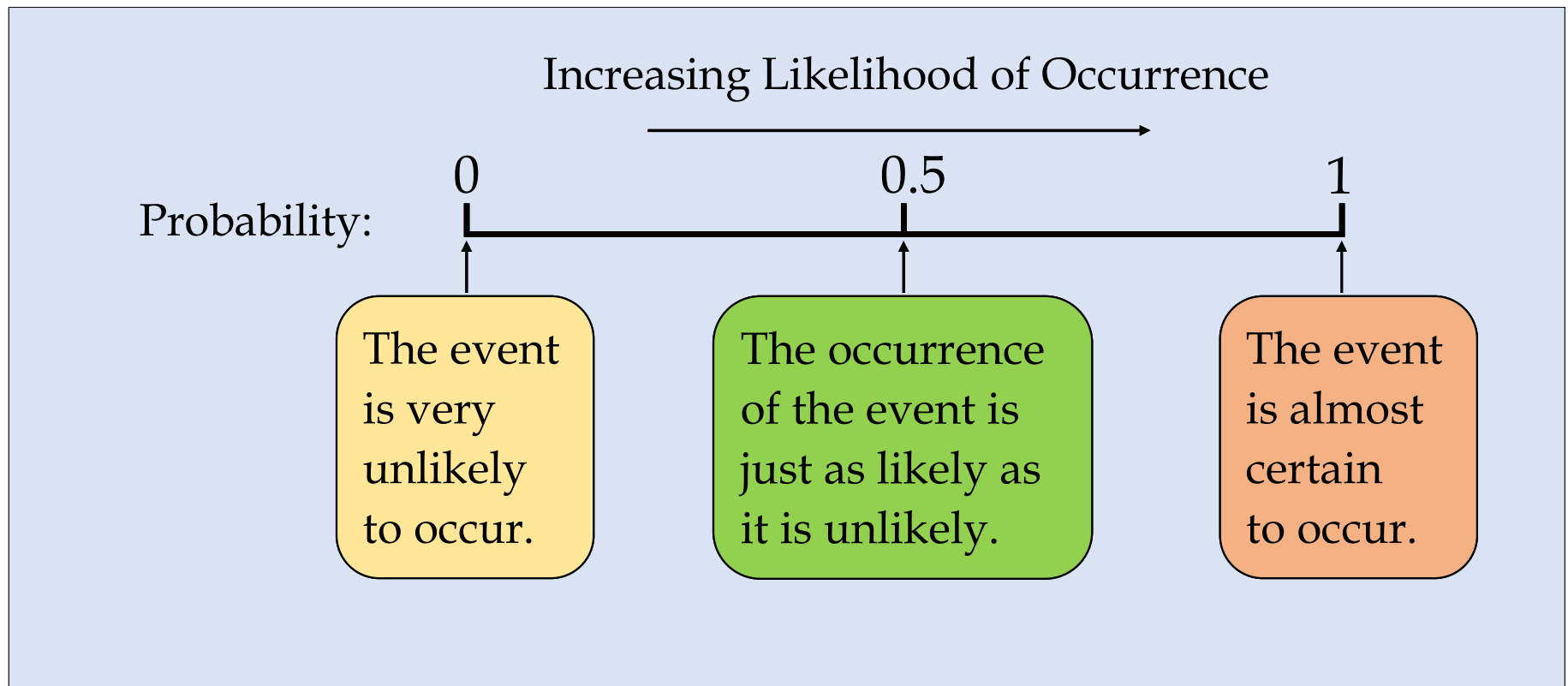
Probability is a numerical measure of the likelihood that an event will occur.

Probability values are always assigned on a scale from 0 to 1.

A probability near zero indicates an event is quite unlikely to occur.

A probability near one indicates an event is almost certain to occur.

Probability as a Numerical Measure of the Likelihood of Occurrence



Statistical Experiments

In statistics, the notion of an experiment differs somewhat from that of an experiment in the physical sciences.

In statistical experiments, probability determines outcomes.

Even though the experiment is repeated in exactly the same way, an entirely different outcome may occur.

For this reason, statistical experiments are some- times called random experiments.

An Experiment and Its Sample Space (S)

An experiment is any process that generates well-defined outcomes.

The sample space(S) for an experiment is the set of all experimental outcomes.

An experimental outcome is also called a sample point.

An Experiment and Its Sample Space

<u>Experiment</u>	<u>Experiment Outcomes</u>	<u>Sample Space</u>
Toss a coin	Head, Tail	$S = \{H, T\}$
Inspection a part	Defective, Non-defective	$S = \{D, ND\}$
Conduct a sales call	Purchase, No purchase	$S = \{P, NP\}$
Roll a die	1,2,3,4,5,6	$S = \{1, 2, 3, 4, 5, 6\}$
Play a football game	Win, Lose, Tie	$S = \{W, L, T\}$

A Counting Rule for Multiple-Step Experiments

If an experiment consists of a sequence of k steps in which there are n_1 possible results for the first step, n_2 possible results for the second step, and so on, then the total number of experimental outcomes is given by $(n_1)(n_2) \dots (n_k)$.

A helpful graphical representation of a multiple-step experiment is a tree diagram

An Experiment and Its Sample Space

Example: Bradley Investments

Bradley has invested in two stocks, Markley Oil and Collins Mining. Bradley has determined that the possible outcomes of these investments three months from now are as follows.

Investment Gain or Loss in 3 Months (in \$000)	
<u>Markley Oil</u>	<u>Collins Mining</u>
10	8
5	-2
0	
-20	

A Counting Rule for Multiple-Step Experiments

Example: Bradley Investments

Bradley Investments can be viewed as a two-step experiment. It involves two stocks, each with a set of experimental outcomes.

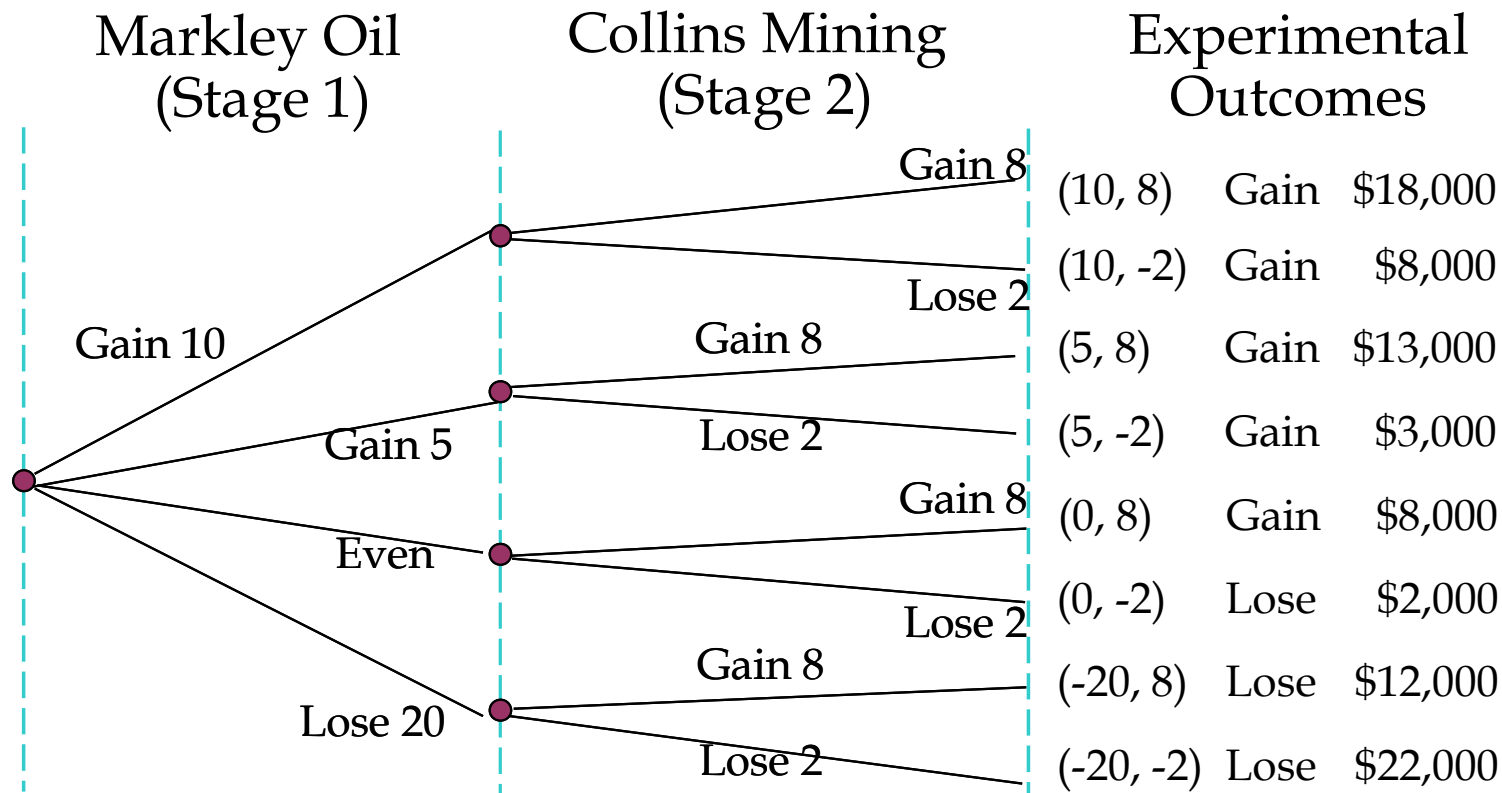
Markley Oil: $n_1 = 4$

Collins Mining: $n_2 = 2$

Total Number of
Experimental Outcomes: $n_1 n_2 = (4)(2) = 8$

Tree Diagram

Example: Bradley Investments



Example: Tossing Coin

Let the experimental outcomes be defined in terms of the pattern of heads and tails appearing on the upward faces of the two coins. How many experimental outcomes are possible for this experiment?

ANS: 2-Step

(1-Step – tossing of the first coin and 2-step –tossing a second coin)

If we use H to denote a head and T to denote a tail, (H, H) indicates the experimental outcome with a head on the first coin and a head on the second coin. Continuing this notation, we can describe the sample space (S) for this coin-tossing experiment as follows:

$$S = \{ (H,H) , (H,T) , (T,H) , (T,T) \}$$

$$n_1 = 2 \text{ \& } n_2 = 2 , \text{ by counting rule } n_1 n_2 = 2 * 2 = 4$$

Tree Diagram

Example: Tossing Coin

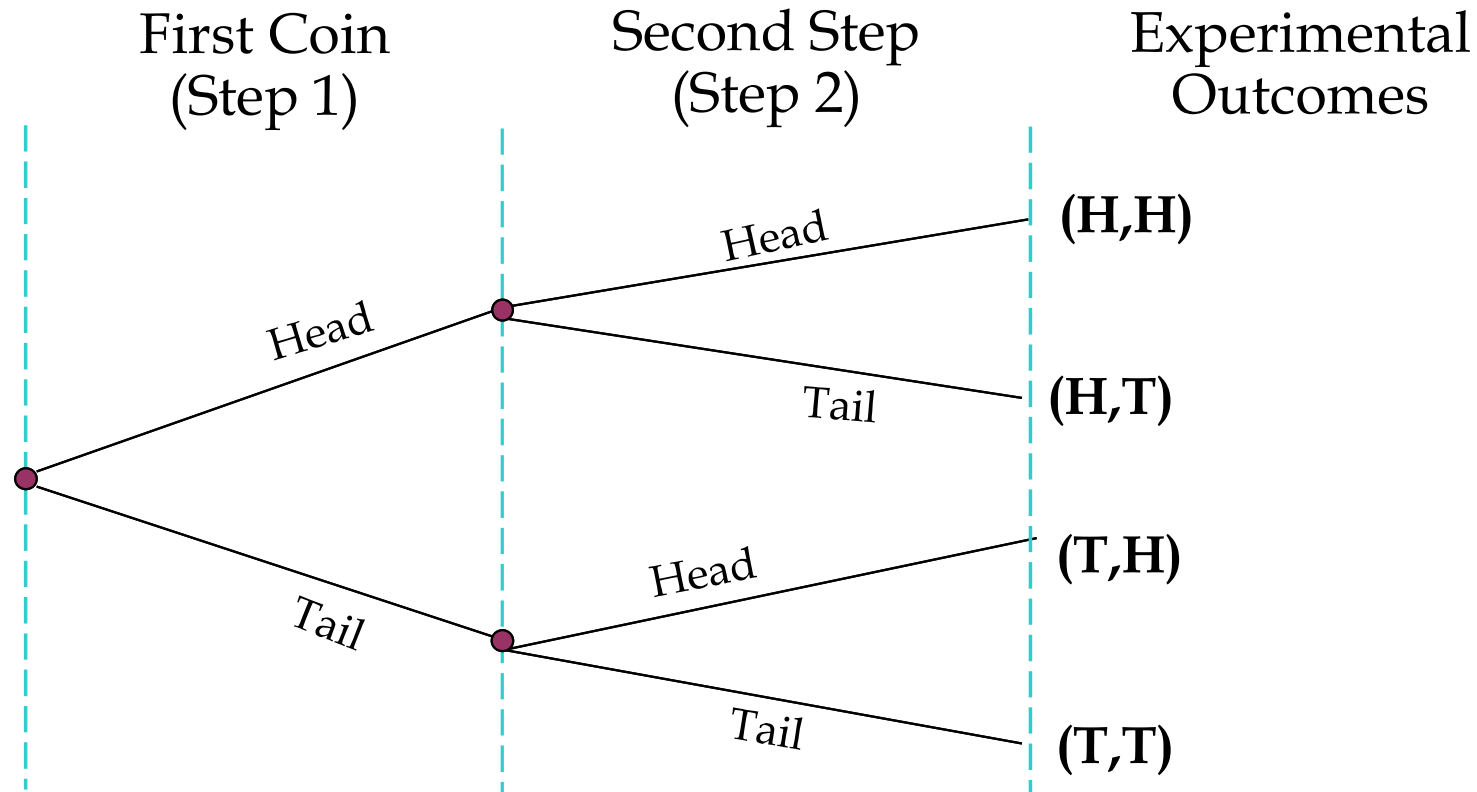
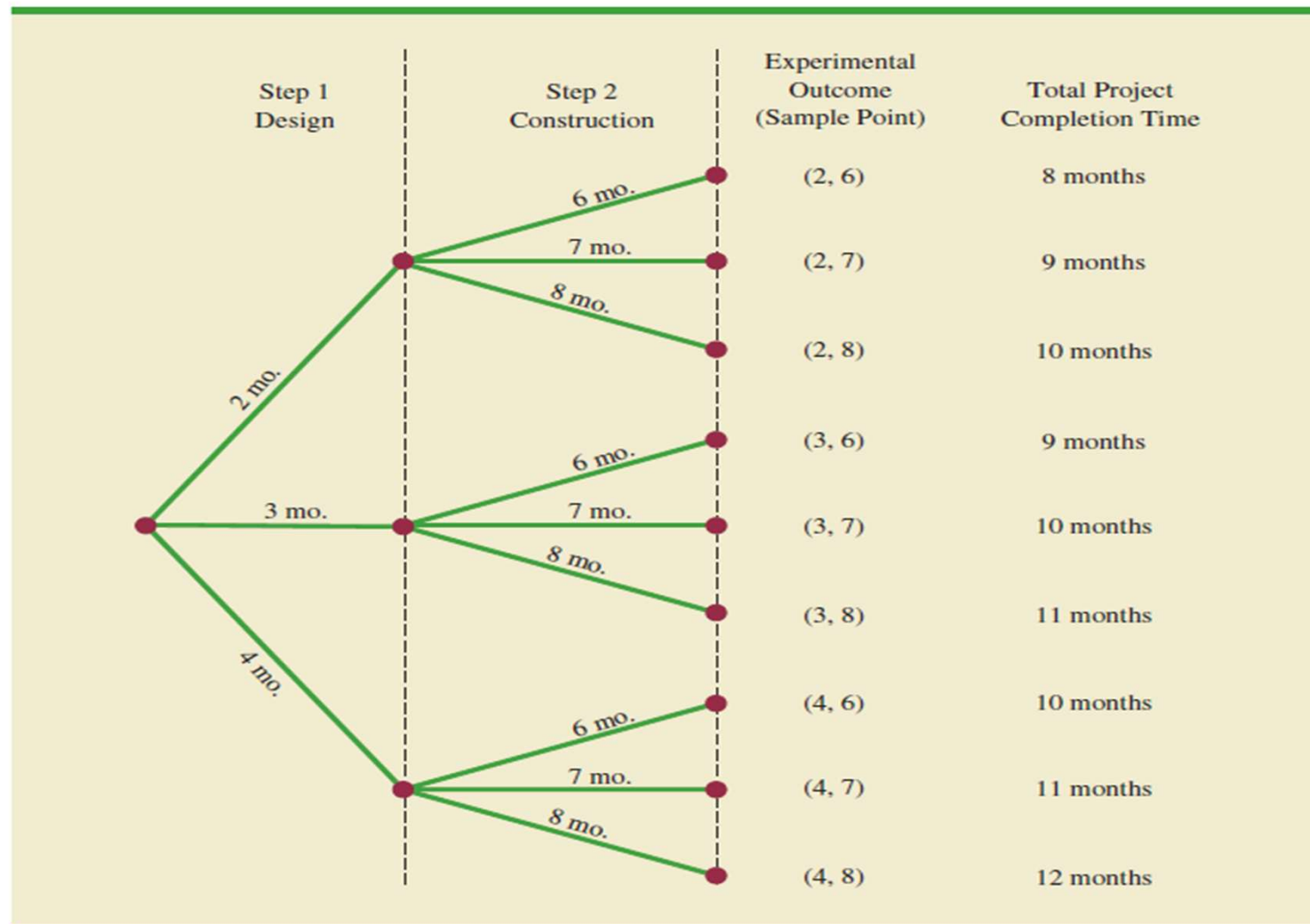


TABLE 4.1 EXPERIMENTAL OUTCOMES (SAMPLE POINTS) FOR THE KP&L PROJECT

Completion Time (months)			
Stage 1 Design	Stage 2 Construction	Notation for Experimental Outcome	Total Project Completion Time (months)
2	6	(2, 6)	8
2	7	(2, 7)	9
2	8	(2, 8)	10
3	6	(3, 6)	9
3	7	(3, 7)	10
3	8	(3, 8)	11
4	6	(4, 6)	10
4	7	(4, 7)	11
4	8	(4, 8)	12

FIGURE 4.3 TREE DIAGRAM FOR THE KP&L PROJECT



Example

1. An experiment has three steps with three outcomes possible for the first step, two outcomes possible for the second step, and four outcomes possible for the third step. How many experimental outcomes exist for the entire experiment?
2. Consider the experiment of tossing a coin three times.
 - a. Develop a tree diagram for the experiment.
 - b. List the experimental outcomes.

Counting Rule for Combinations

Number of Combinations of N Objects Taken n at a Time

A second useful counting rule enables us to count the number of experimental outcomes when n objects are to be selected from a set of N objects.

$$C_n^N = \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

where: $N! = N(N-1)(N-2) \dots (2)(1)$
 $n! = n(n-1)(n-2) \dots (2)(1)$
 $0! = 1$

Example

Consider a quality control procedure in which an inspector randomly selects two of five parts to test for defects. In a group of five parts, how many combinations of two parts can be selected?

$N = 5$ and $n = 2$

$$C_2^5 = \binom{5}{2} = \frac{5!}{2!(5-2)!} = \frac{(5)(4)(3)(2)(1)}{(2)(1)(3)(2)(1)} = \frac{120}{12} = 10$$

10 outcomes are possible for the experiment of randomly selecting two parts from a group of five. If we label the five parts as A, B, C, D, and E, the 10 combinations or experimental outcomes can be identified as AB, AC, AD, AE, BC, BD, BE, CD, CE, and DE.

Example

Consider that the Florida lottery system uses the random selection of six integers from a group of 53 to determine the weekly winner.

$$\binom{53}{6} = \frac{53!}{6!(53-6)!} = \frac{53!}{6!47!} = \frac{(53)(52)(51)(50)(49)(48)}{(6)(5)(4)(3)(2)(1)} = 22,957,480$$

Counting Rule for Permutations

Number of Permutations of N Objects Taken n at a Time

A third useful counting rule enables us to count the number of experimental outcomes when n objects are to be selected from a set of N objects. where the order of selection is important.

$$P_n^N = n! \binom{N}{n} = \frac{N!}{(N-n)!}$$

where: $N! = N(N-1)(N-2) \dots (2)(1)$

$n! = n(n-1)(n-2) \dots (2)(1)$

$0! = 1$

Example

Consider again the quality control process in which an inspector selects two of five parts to inspect for defects. How many permutations may be selected?

$N = 5$ and $n = 2$

$$P_2^5 = \frac{5!}{(5 - 2)!} = \frac{5!}{3!} = \frac{(5)(4)(3)(2)(1)}{(3)(2)(1)} = \frac{120}{6} = 20$$

20 outcomes are possible for the experiment of randomly selecting two parts from a group of five when the order of selection must be taken into account. If we label the parts A, B, C, D, and E, the 20 permutations are AB, BA, AC, CA, AD, DA, AE, EA, BC, CB, BD, DB, BE, EB, CD, DC, CE, EC, DE, and ED.

Example

1. How many ways can three items be selected from a group of six items? Use the letters A, B, C, D, E, and F to identify the items, and list each of the different combinations of three items.
2. How many permutations of three items can be selected from a group of six? Use the letters A, B, C, D, E, and F to identify the items, and list each of the permutations of items B, D, and F.

Difference between Permutations and Combinations

1. A permutation is an act of arranging objects or numbers in order. Permutations are used when order/sequence of arrangement is needed.
2. Combinations are the way of selecting objects or numbers from a group of objects or collections, in such a way that the order of the objects does not matter. Combinations are used to find the number of possible groups which can be formed.

Assigning Probabilities

Basic Requirements for Assigning Probabilities

1. The probability assigned to each experimental outcome must be between 0 and 1, inclusively.

$$0 \leq P(E_i) \leq 1 \text{ for all } i$$

where: E_i is the i^{th} experimental outcome and
 $P(E_i)$ is its probability

Assigning Probabilities

Basic Requirements for Assigning Probabilities

2. The sum of the probabilities for all experimental outcomes must equal 1.

$$P(E_1) + P(E_2) + \dots + P(E_n) = 1$$

where: n is the number of experimental outcomes

Assigning Probabilities

1. Classical Method

Assigning probabilities based on the assumption of equally likely outcomes

2. Relative Frequency Method

Assigning probabilities based on experimentation or historical data

3. Subjective Method

Assigning probabilities based on judgment

1. Classical Method

Example: Rolling a Die

If an experiment has n possible outcomes, the classical method would assign a probability of $1/n$ to each outcome.

Experiment: Rolling a die

Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$

Probabilities: Each sample point has a $1/6$ chance of occurring.

2. Relative frequency Method

Example: ABC Rental

ABC Rental would like to assign probabilities to the number of car polishers it rents each day. Office records show the following frequencies of daily rentals for the last 40 days.

<u>Number of Polishers Rented</u>	<u>Number of Days</u>
0	4
1	6
2	18
3	10
4	2

2. Relative frequency Method

Example: ABC Rental

Each probability assignment is given by dividing the frequency (number of days) by the total frequency (total number of days).

<u>Number of Polishers Rented</u>	<u>Number of Days</u>	<u>Probability</u>
0	4	.10
1	6	.15
2	18	.45
3	10	.25
4	<u>2</u>	<u>.05</u>
	40	1.00



4/40

3. Subjective Method

When economic conditions and a company's circumstances change rapidly it might be inappropriate to assign probabilities based solely on historical data.

We can use any data available as well as our experience and intuition, but ultimately a probability value should express our degree of belief that the experimental outcome will occur.

The best probability estimates often are obtained by combining the estimates from the classical or relative frequency approach with the subjective estimate.

3. Subjective Method

Example: Bradley Investments

An analyst made the following probability estimates.

<u>Exper. Outcome</u>	<u>Net Gain or Loss</u>	<u>Probability</u>
(10, 8)	\$18,000 Gain	.20
(10, -2)	\$8,000 Gain	.08
(5, 8)	\$13,000 Gain	.16
(5, -2)	\$3,000 Gain	.26
(0, 8)	\$8,000 Gain	.10
(0, -2)	\$2,000 Loss	.12
(-20, 8)	\$12,000 Loss	.02
(-20, -2)	\$22,000 Loss	.06

Example : KP&L Project

On the basis of experience and judgment, management concluded that the experimental outcomes were not equally likely. Hence, **the classical method of assigning probabilities could not be used**. Management then decided to conduct a study of the completion times for similar projects undertaken by KP&L over the past three years. The results of a study of 40 similar projects are summarized in Table 4.2. Management could have provided subjective probability estimates, but felt that the current project was quite similar to the 40 previous projects. Thus, **the relative frequency method was judged best**.

TABLE 4.2 COMPLETION RESULTS FOR 40 KP&L PROJECTS

Completion Time (months)		Sample Point	Number of Past Projects Having These Completion Times
Stage 1 Design	Stage 2 Construction		
2	6	(2, 6)	6
2	7	(2, 7)	6
2	8	(2, 8)	2
3	6	(3, 6)	4
3	7	(3, 7)	8
3	8	(3, 8)	2
4	6	(4, 6)	2
4	7	(4, 7)	4
4	8	(4, 8)	6
Total			40

Example : KP&L Project

TABLE 4.3 PROBABILITY ASSIGNMENTS FOR THE KP&L PROJECT BASED ON THE RELATIVE FREQUENCY METHOD

Sample Point	Project Completion Time	Probability of Sample Point
(2, 6)	8 months	$P(2, 6) = 6/40 = .15$
(2, 7)	9 months	$P(2, 7) = 6/40 = .15$
(2, 8)	10 months	$P(2, 8) = 2/40 = .05$
(3, 6)	9 months	$P(3, 6) = 4/40 = .10$
(3, 7)	10 months	$P(3, 7) = 8/40 = .20$
(3, 8)	11 months	$P(3, 8) = 2/40 = .05$
(4, 6)	10 months	$P(4, 6) = 2/40 = .05$
(4, 7)	11 months	$P(4, 7) = 4/40 = .10$
(4, 8)	12 months	$P(4, 8) = 6/40 = .15$
		Total 1.00

Example

1. Consider the experiment of tossing a coin three times.
What is the probability for each experimental outcome? (4)
2. An experiment with three outcomes has been repeated 50 times, and it was learned that E_1 occurred 20 times, E_2 occurred 13 times, and E_3 occurred 17 times. Assign probabilities to the outcomes.
What is the probability for each experimental outcome? What method did you use? (6)
3. A decision maker subjectively assigned the following probabilities to the four outcomes of an experiment: $P(E_1) = 0.10$, $P(E_2) = 0.15$, $P(E_3) = .40$, and $P(E_4) = .20$. Are these probability assignments valid? Explain. (7)

Example

4. Suppose an experiment has five equally likely outcomes: E1, E2, E3, E4, E5. Assign probabilities to each outcome. What method did you use? (5)
5. Simple random sampling uses a sample of size n from a population of size N to obtain data that can be used to make inferences about the characteristics of a population. Suppose that, from a population of 50 bank accounts, we want to take a random sample of four accounts in order to learn about the population. How many different random samples of four accounts are possible? (9)

Events and Their Probabilities

An event is a collection of sample points.

The probability of any event is equal to the sum of the probabilities of the sample points in the event.

If we can identify all the sample points of an experiment and assign a probability to each, we can compute the probability of an event.

Example: Bradley Investments

An analyst made the following probability estimates.

<u>Exper. Outcome</u>	<u>Net Gain or Loss</u>	<u>Probability</u>
(10, 8)	\$18,000 Gain	.20
(10, -2)	\$8,000 Gain	.08
(5, 8)	\$13,000 Gain	.16
(5, -2)	\$3,000 Gain	.26
(0, 8)	\$8,000 Gain	.10
(0, -2)	\$2,000 Loss	.12
(-20, 8)	\$12,000 Loss	.02
(-20, -2)	\$22,000 Loss	.06

Events and Their Probabilities

Example: Bradley Investments

Event M = Markley Oil Profitable

$$M = \{(10, 8), (10, -2), (5, 8), (5, -2)\}$$

$$P(M) = P(10, 8) + P(10, -2) + P(5, 8) + P(5, -2)$$

$$= .20 + .08 + .16 + .26$$

$$= .70$$

Events and Their Probabilities

Example: Bradley Investments

Event C = Collins Mining Profitable

$$C = \{(10, 8), (5, 8), (0, 8), (-20, 8)\}$$

$$P(C) = P(10, 8) + P(5, 8) + P(0, 8) + P(-20, 8)$$

$$= .20 + .16 + .10 + .02$$

$$= .48$$

Example

1. An experiment has four equally likely outcomes: E_1, E_2, E_3 , and E_4 . (14)
 - a. What is the probability that E_2 occurs?
 - b. What is the probability that any two of the outcomes occur (e.g., E_1 or E_3)?
 - c. What is the probability that any three of the outcomes occur (e.g., E_1 or E_2 or E_4)?
2. Consider the experiment of rolling a pair of dice. Suppose that we are interested in the sum of the face values showing on the dice. (16)
 - a. How many sample points are possible? List the sample points.
 - b. What is the probability of obtaining a value of 7?
 - c. What is the probability of obtaining a value of 9 or greater?
 - d. Because each roll has six possible even values (2, 4, 6, 8, 10, and 12) and only five possible odd values (3, 5, 7, 9, and 11), the dice should show even values more often than odd values. Do you agree with this statement? Explain.
 - e. What method did you use to assign the probabilities requested?

Example

1. The sample space of a random experiment of selecting a number is $U = \{1, 2, 3, \dots, 20\}$. Write the sets and probability for the following events:
 - a. The selected number is odd number.
 - b. The selected number is divisible by 3.
 - c. The selected number is divisible by 2 or 3.
2. One family is selected from the families having two children. The male or female of the children from this family is noted. State the sample space of this experiment and write the sets showing the following events:
 - a. Event A_1 = One child is a female
 - b. Event A_2 = At least one child is a female
 - c. Event A_3 = One child is a male
 - d. Event A_4 = At least one child is a male

Some Basic Relationships of Probability

There are some basic probability relationships that can be used to compute the probability of an event without knowledge of all the sample point probabilities.

Complement of an Event

Union of Two Events

Intersection of Two Events

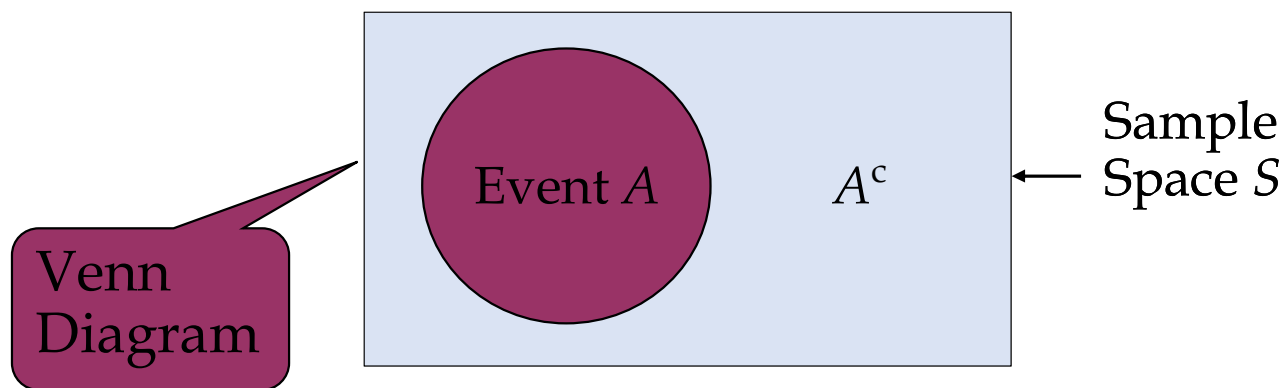
Mutually Exclusive Events

Complement of an Event

The complement of event A is defined to be the event consisting of all sample points that are not in A .

The complement of A is denoted by A^c .

Computing Prob. using complement : $P(A) = 1 - P(A^c)$



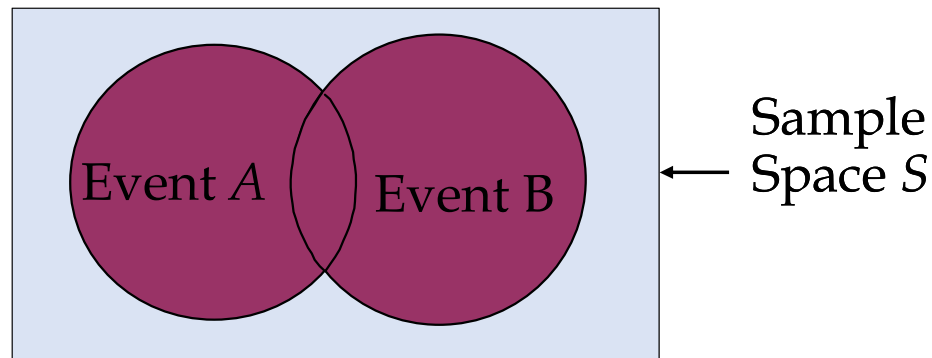
Example

1. Two events connected with the experiment of rolling a single die are E : “the number rolled is even” and T : “the number rolled is greater than two.” Find the complement of each.
2. Suppose you know that the probability of getting the flu this winter is 0.43. What is the probability that you will not get the flu?
3. Two coins are tossed simultaneously. Let the event A be observing at least one head. What is the complement of A , and how would you calculate the probability of A by using the Complement Rule?

Union of Two Events

The union of events A and B is the event containing all sample points that are in A or B or both.

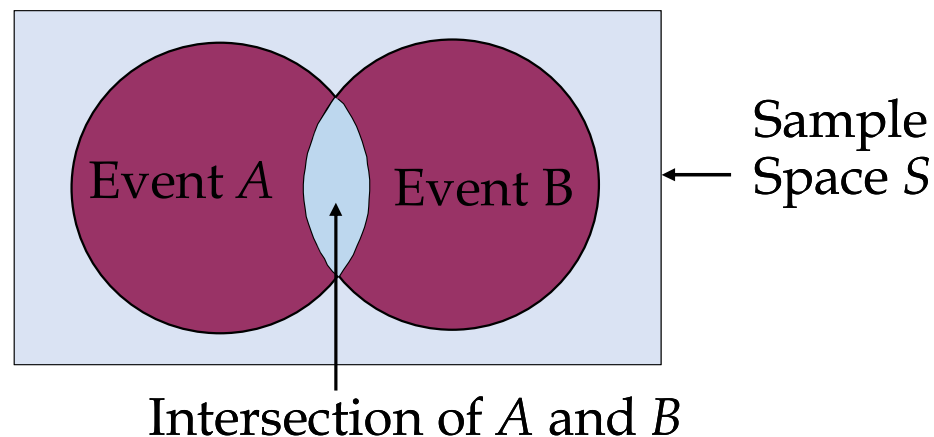
The union of events A and B is denoted by $A \cup B$.



Intersection of Two Events

The intersection of events A and B is the set of all sample points that are in both A and B .

The intersection of events A and B is denoted by $A \cap B$.



Example

1. If $A = \{1,3,5\}$ and $B = \{1,2,4\}$ then $A \cup B = ?$
2. If $A = \{1,3,5\}$ and $B = \{1,2,4\}$ then $A \cap B = ?$
3. Two numbers are selected at random from the first five natural numbers. The sum of two selected numbers is at least 6 is denoted by event A and the sum of two selected numbers is even is denoted by event B. Write the sets showing the following events and answer the given questions: The sum of two selected numbers is at least 6 or even, The sum of two selected numbers is at least 6 and even, The sum of two selected numbers is at the most 6.

Addition Law

The addition law provides a way to compute the probability of Event A or B or Both A and B occurring.

The law is written as:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

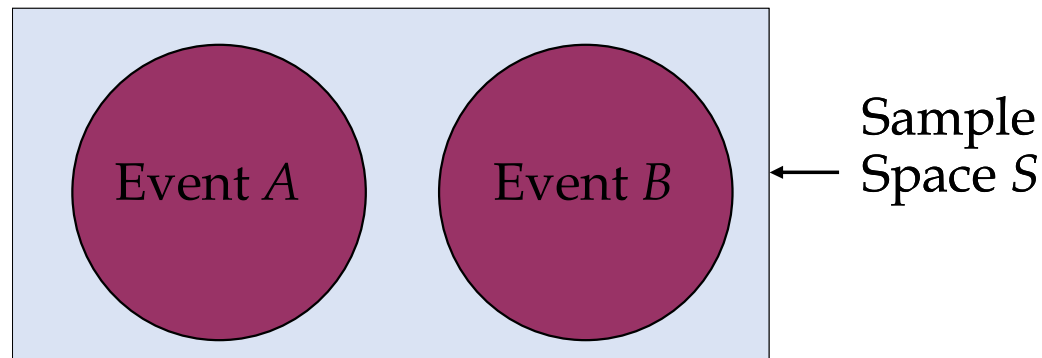
Example

1. let us consider the case of a small assembly plant with 50 employees. Each worker is expected to complete work assignments on time and in such a way that the assembled product will pass a final inspection. On occasion, some of the workers fail to meet the performance standards by completing work late or assembling a defective product. At the end of a performance evaluation period, the production manager found that 5 of the 50 workers completed work late, 6 of the 50 workers assembled a defective product, and 2 of the 50 workers both completed work late and assembled a defective product. What is the probability that the production manager assigned an employee a poor performance rating?

Mutually Exclusive Events

Two events are said to be Mutually Exclusive if the events have no sample points in common.

Two events are mutually exclusive if, **when one event occurs, the other cannot occur.**



Mutually Exclusive Events

If events A and B are mutually exclusive, $P(A \cap B) = 0$.

The addition law for mutually exclusive events is:

$$P(A \cup B) = P(A) + P(B)$$

There is no need to include " $- P(A \cap B)$ "

Example

1. Suppose that we have a sample space with five equally likely experimental outcomes: E_1, E_2, E_3, E_4, E_5 . Let $A = \{E_1, E_2\}$, $B = \{E_3, E_4\}$, $C = \{E_2, E_3, E_5\}$ (22)
 - a. Find $P(A)$, $P(B)$, and $P(C)$.
 - b. Find $A \cup B$, $A \cap B$ and $P(A \cup B)$, $P(A \cap B)$.
 - c. Are events A and C mutually exclusive?
 - d. Find A^c , C^c , $A \cup B^c$, $P(A^c)$, $P(C^c)$ and $P(A \cup B^c)$.
2. (25) The U.S. Census Bureau provides data on the number of young adults, ages 18–24, who are living in their parents' home. Let
 M = the event a male young adult is living in his parents' home
 F = the event a female young adult is living in her parents' home
If we randomly select a male young adult and a female young adult, the Census Bureau data enable us to conclude $P(M) = 0.56$ and $P(F) = 0.42$.
The probability that both are living in their parents' home is .24.

Example

- a. What is the probability at least one of the two young adults selected is living in his or her parents' home?
 - b. What is the probability both young adults selected are living on their own (neither is living in their parents' home)?
3. (28) A survey of magazine subscribers showed that 45.8% rented a car during the past 12 months for business reasons, 54% rented a car during the past 12 months for personal reasons, and 30% rented a car during the past 12 months for both business and personal reasons.
 - a. What is the probability that a subscriber rented a car during the past 12 months for business or personal reasons?
 - b. What is the probability that a subscriber did not rent a car during the past 12 months for either business or personal reasons?

Conditional Probability

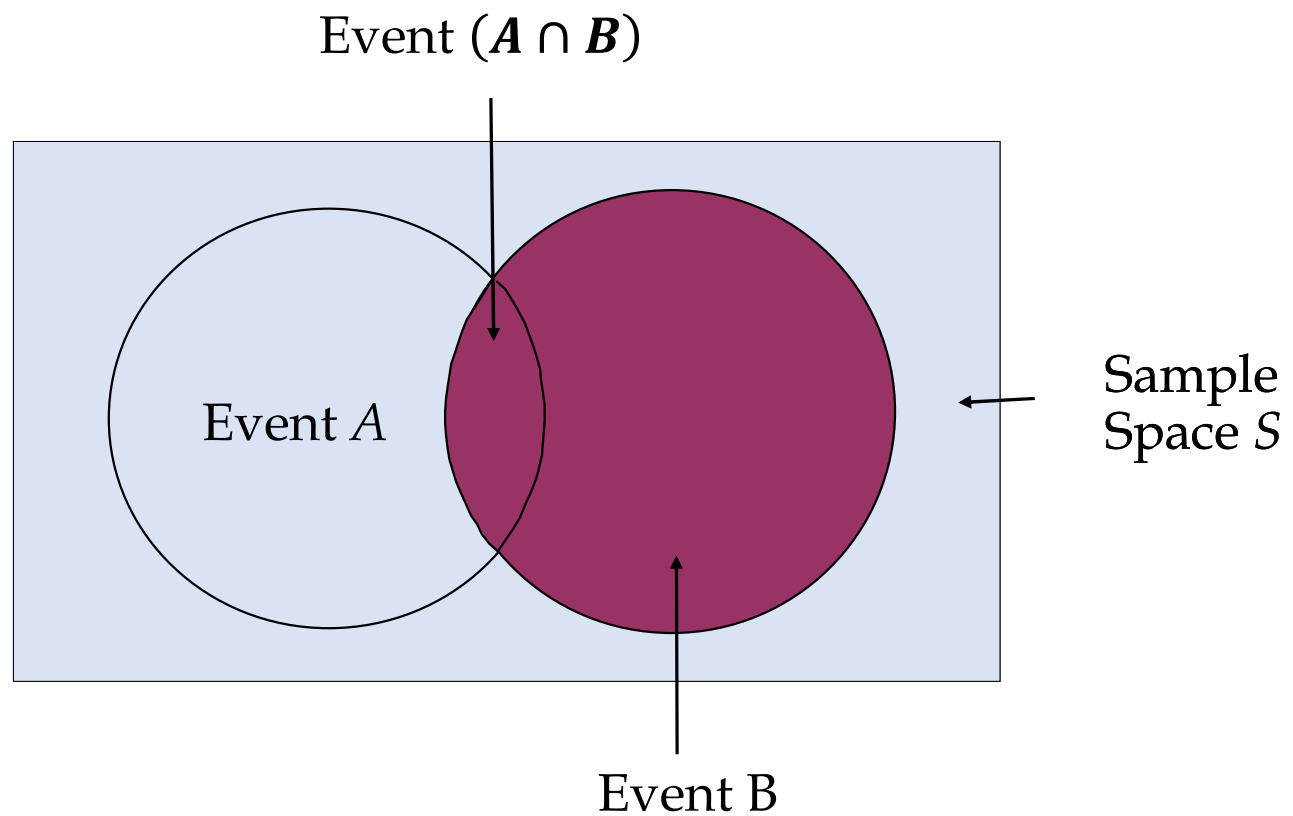
The probability of an even given that another event has Occurred is called a conditional probability.

The conditional probability of A given B is denoted by $P(A | B)$.

A conditional probability is computed as follows:

$$P(A | B) = P \frac{(A \cap B)}{P(B)}$$

Conditional Probability



Conditional Probability

Example: Consider the situation of the promotion status of male and female officers of a major metropolitan police force in the eastern United States. The police force consists of 1200 officers, 960 men and 240 women. Over the past two years, 324 officers on the police force received promotions. After reviewing the promotion record, a committee of female officers raised a discrimination case on the basis that 288 male officers had received promotions but only 36 female officers had received promotions. The police administration argued that the relatively low number of promotions for female officers was due not to discrimination, but to the fact that relatively few females are members of the police force.

Conditional Probability

	Men (M)	Women (W)	Total
Promoted (A)	288	36	324
Not Promoted (A^c)	672	204	876
Total	960	240	1200

Conditional Probability

M = event an officer is a man

W = event an officer is a woman

A = event an officer is promoted

A^c = event an officer is not promoted

$$P(M \cap A) = 288/1200 = 0.24$$

= probability that a randomly selected officer
is a man and is promoted

$$P(M \cap A^c) = 672/1200 = 0.56$$

= probability that a randomly selected officer
is a man and is not promoted

Conditional Probability

$$P(W \cap A) = 36/1200 = 0.03$$

= probability that a randomly selected officer
is a woman and is promoted

$$P(W \cap A^c) = 204/1200 = 0.17$$

= probability that a randomly selected officer
is a woman and is not promoted

Conditional Probability

	Men (M)	Women (W)	Total
Promoted (A)	0.24	0.03	0.27
Not Promoted (A^c)	0.56	0.17	0.73
Total	0.80	0.20	1.00

Joint Probabilities

Marginal Probabilities

Joint Probability: Because each of these values gives the probability of the intersection of two events, the probabilities are called joint probabilities.

Marginal Probability: The values in the margins of the joint probability table provide the probabilities of each event separately. For example,

$$P(M) = 0.80 , P(W) = 0.20 , P(A) = 0.27 , P(A^c) = 0.73$$

Marginal Probability:

the marginal probabilities are found by summing the joint probabilities in the corresponding row or column of the joint Probability table. For instance, the marginal probability of being promoted is

$$\begin{aligned} P(A) &= P(M \cap A) + P(W \cap A) \\ &= 0.24 + 0.03 = 0.27 \end{aligned}$$

From the marginal probabilities, we see that 80% of the force is male, 20% of the force is female, 27% of all officers received promotions, and 73% were not promoted.

Conditional Probability: The probability that an officer is promoted given that the officer is a man.

Notation : $P(A|M)$

Description :

$P(A|M)$ = we are now concerned only with the promotion status of the 960 male officers and 288 of the 960 male officers received Promotions. The probability of being promoted given that the officer is a man is $288/960 = 0.30$. In other words, given that an officer is a man, that officer had a 30% chance of receiving a Promotion over the past two years.

$$P(A|M) = \frac{288}{960} = \frac{288/1200}{960/1200} = 0.30$$

$$P(A|M) = P \frac{(A \cap M)}{P(M)} = \frac{0.24}{0.80} = 0.30$$

Find $P(A|W) = ?$

Ans : 0.15

Independent Events

Two events A and B are independent if

$$P(A \mid B) = P(A)$$

Or

$$P(B \mid A) = P(B)$$

Otherwise, the events are dependent.

Multiplication Law

The **multiplication law** provides a way to compute the probability of the intersection of two events.

The law is written as:

$$P(A \cap B) = P(B) P(A | B)$$

Or

$$P(A \cap B) = P(A) P(B | A)$$

Multiplication Law for Independent Events

The law is written as:

$$P(A \cap B) = P(B) P(A)$$

Example

1. If A and B are two events such that $P(A) = 2/3$, $P(B) = 1/2$ and $P(A \cap B) = 1/3$. Find $P(A \cup B)$, $P(A|B)$, $P(B|A)$. And explain whether the events A and B are mutually exclusive and independent?
2. Suppose that we have two events, A and B , with $P(A) = .50$, $P(B) = .60$, and $P(A \cap B) = .40$. (30)
 - a) Find $P(A|B)$
 - b) Find $P(B|A)$
 - c) Are A and B independent? Why or why not?

3. (32)The automobile industry sold 657,000 vehicles in the United States during January 2009 (The Wall Street Journal, February 4, 2009). This volume was down 37% from January 2008 as economic conditions continued to decline. A summary of sales by automobile manufacturer and type of vehicle sold is shown in the following table. Data are in thousands of vehicles. The non-U.S. manufacturers are led by Toyota, Honda, and Nissan. The category Light Truck includes pickup, minivan, SUV, and crossover models.

Manufacturer		Types of Vehicle	
		Car	Light Truck
	U.S.	87.4	193.1
	Non U.S.	228.5	148.0

- a) Develop a joint probability table for these data and use the table to answer the remaining questions.
- b) What are the marginal probabilities? What do they tell you about the probabilities associated with the manufacturer and the type of vehicle sold?
- c) If a vehicle was manufactured by one of the U.S. automakers, what is the probability that the vehicle was a car? What is the probability it was a light truck?
- d) If a vehicle was not manufactured by one of the U.S. automakers, what is the probability that the vehicle was a car? What is the probability it was a light truck?
- e) If the vehicle was a light truck, what is the probability that it was manufactured by one of the U.S. automakers?
- f) What does the probability information tell you about sales?

Probability and Probability distribution

Probability distribution



Random Variables with its types

Random Variables

In probability theory and statistics, a random variable is a variable that takes on different values based on the outcomes of a random experiment or a probability distribution.

Random variables provide a way to formalize and quantify uncertainty in a mathematical framework. They are crucial for modeling and analyzing various phenomena in probabilistic settings.

There are two main types of random variables:

1. Discrete Random Variables
2. Continuous Random Variables

In some cases, random variables can be a combination of both discrete and continuous types, referred to as mixed random variables.

Random variables are fundamental in probability theory and statistics, and they play a central role in various statistical analyses, hypothesis testing, and probabilistic modeling. They provide a formal and mathematical framework for understanding and quantifying the uncertainty inherent in many real-world phenomena.



Random Variables with its types

Discrete Random Variables

A discrete random variable is a random variable that takes on a countable set of distinct values, meaning its possible values can be listed one by one. These values are often represented as $x_1, x_2, x_3, \dots, x_n$ where n is the number of distinct values the random variable can take.

Discrete random variables have a countable number of possible outcomes. This means that the set of possible values can be listed, and there are gaps between the values.

Moreover, Discrete random variable has small range of outcome but more repetition in the outcome can be observed.

Example:

1. Number of cars, family members, houses, people or some other countable sets.
2. Morel values, number of rules and regulation, number of outcomes observed after rolling a die or two or more than two dies.
3. Outcomes of any indoor or outdoor games, any dichotomous variables, outcomes of any real-life arguments.
4. Any denumerable set or countably infinite set



Continuous Random Variables

A continuous random variable is a type of random variable that can take on any value within a certain range or interval. The values are often represented by open, close or sometimes clopen intervals.

Unlike discrete random variables, which have isolated and countable possible values, continuous random variables have an uncountably infinite number of potential outcomes within their range. There is no gap in between for continuous random variables.

Moreover, Continuous random variable has huge range of outcome but very less repetition in the outcome can be observed.

Example:

1. Age of population, Reaction time, Wind speed, Rainfall amount, Blood pressure, Temperature Change or some other uncountable physical quantities.
2. Income level, Height of plants, Lifetime of electronic devices, Lifespan of Uranium or other atoms, Waiting time in a queue.
3. In financial instruments like Stock price, Volatility, Fluctuations in stock indices, Income of company, option pricing etc.
4. Any uncountable infinite set



Probability Functions

Probability Functions

A probability function, also known as a probability distribution function, assigns probabilities to different outcomes of a random variable. It describes how the probabilities are distributed across the possible values of the random variable.

The specific type of probability function depends on whether the random variable is discrete or continuous and in both type of random variables, cumulative distribution can always be calculated to find cumulative spread.

There are mainly three types of probability distribution functions

1. Probability Mass Function (PMF)
2. Probability Density Function (PDF)
3. Cumulative Distribution Function (CDF)



Probability Mass Functions

The Probability Mass Function (PMF) is a fundamental concept in probability theory and is used to describe the probability distribution of a discrete random variable. It gives the probability of the random variable taking on specific values in its domain.

For a discrete random variable X , the PMF is denoted as $P(X = x)$, where x represents a particular value in the domain of X . The PMF provides the probability of X being equal to that specific value x .

Properties of the PMF:

1. Non-Negativity: The PMF is non-negative for all values of x : $P(X = x) \geq 0$ for all x in the domain of X .
2. Probability Sum: The sum of the probabilities for all possible values of X is equal to 1

$$\sum P(X = x) = 1$$

where the summation is taken over all possible values of X .

Example: Tossing a Fair Coin

Let's consider a simple example of a discrete random variable X representing the outcome of tossing a fair coin once. The possible values of X are "Heads" (H) and "Tails" (T).

PMF for both outcomes is 0.5 here.



Probability Density Functions

The Probability Density Function (PDF) is a concept in probability theory used to describe the probability distribution of a continuous random variable. Unlike the Probability Mass Function (PMF) for discrete random variables, the PDF does not give the probability of the random variable taking on specific values.

Instead, it provides the relative likelihood of the random variable falling within a particular range of values.

For a continuous random variable X , the PDF is denoted as $f(x)$, where x represents a specific value in the domain of X . The PDF provides the rate at which the probability changes with respect to x . In other words, the probability that X falls within a particular interval $[a, b]$ is given by the integral of the PDF over that interval:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Properties of the PDF:

1. Non-Negativity: The PDF is non-negative for all values of x : $f(x) \geq 0$ for all x in the domain of X .
2. Total Area under the Curve:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



Probability Density Functions

The PDF provides the foundation for performing various probabilistic calculations and statistical analyses involving continuous random variables. It allows us to understand the likelihood of different ranges of outcomes and make inferences about the behavior of the random variable.

Example: Consider a continuous random variable X representing the height of students in a classroom. Let's assume that the heights of students in this class can vary continuously between a minimum height of 140 cm and a maximum height of 180 cm.

In this example, the PDF is a function that describes the relative likelihood of a student having a particular height within the range $[140, 180]$.

PDF Characteristics:

1. The PDF is a smooth curve over the interval $[140, 180]$ since the heights can vary continuously.
2. The PDF is non-negative, meaning the probability of a student having a specific height is always non-negative.
3. The total area under the curve of the PDF over the entire range of heights (from 140 to 180 cm) is equal to 1, indicating that the probability of a student's height falling within this entire range is 100%.



Cumulative Distribution Functions

The Cumulative Distribution Function (CDF) is a concept in probability theory used to describe the probability distribution of both discrete and continuous random variables. The CDF gives the probability that a random variable takes on a value less than or equal to a specified value. It provides a cumulative view of the probability distribution, showing how probabilities accumulate as the value of the random variable increases.

Discrete Random Variable:

For a discrete random variable X , the CDF is denoted as $F(X = x)$ or simply $F(x)$. It is defined as the sum of the probabilities of all values less than or equal to x :

$$F(x) = P(X \leq x) = \sum P(X = x_i) \text{ for all } x_i \leq x.$$

Properties of the CDF for Discrete Random Variables:

1. The CDF is non-decreasing: As x increases, the value of $F(x)$ does not decrease; it only increases or remains constant.
2. The CDF is bounded: For any discrete random variable X , $0 \leq F(x) \leq 1$ for all x .
3. The CDF is right-continuous: The CDF has no jumps; it remains continuous from the right.



Cumulative Distribution Functions

Continuous Random Variable:

For a continuous random variable X , the CDF is denoted as $F(x)$. It is defined as the integral of the probability density function (PDF) up to the value x :

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

where $f(t)$ is the PDF of the continuous random variable X .

Properties of the CDF for Continuous Random Variables:

1. The CDF is non-decreasing: As x increases, the value of $F(x)$ does not decrease; it only increases or remains constant.
2. The CDF is bounded: For any continuous random variable X , $0 \leq F(x) \leq 1$ for all x .
3. The CDF is right-continuous: The CDF is continuous from the right, but it may have discontinuities (jumps) at certain points where the random variable has a non-zero probability.
4. The CDF is differentiable: If the PDF of X is continuous, then the CDF is differentiable almost everywhere.
5. The CDF provides a valuable tool for various probabilistic calculations and statistical analyses, including finding probabilities for specific intervals, computing percentiles, and determining the expected value and variance of a random variable.



Moments:

In statistics, moments are a set of quantitative measures that provide information about the shape, central tendency, and variability of a probability distribution or a data set. Moments are calculated based on the values of a random variable or data points and their corresponding probabilities or frequencies.

Mean of x , x^2 , x^3 and x^4 are called first, second, third and fourth moments or raw moments.

Denoted as: $E(x)$, $E(x^2)$, $E(x^3)$ and $E(x^4)$

Here we should understand that moments or raw moments can only give information about central tendency.

Central Moment:

The central moments are a special type of moments that are centered around the mean of the distribution. They provide information about the dispersion and shape of the distribution without being influenced by the location of the distribution's center.

First central moment is zero: $E(x - \mu) = 0$

Second central moment is variance: $\text{Variance}(x) = E(x - \mu)^2$

Third moment is Skewness: $\text{Skewness}(x) = \frac{E(x - \mu)^3}{(\text{var}(x))^3}$

Fourth moment is Kurtosis: $\text{Kurtosis}(x) = \frac{E(x - \mu)^4}{(\text{var}(x))^4}$



Expectation (Arithmetic Mean/ Average/ First Moment)

The expectation (E) or mean of a random variable X is a measure of its central tendency and represents the average value of the random variable.

For a discrete random variable X with probability mass function (PMF) $P(X = x)$, the expectation is given by:

$$E(X) = \sum (x * P(X = x)) \quad \forall x > 0$$

For a continuous random variable X with probability density function (PDF) $f(x)$, the expectation is given by:

$$E(X) = \int (x * f(x)) dx \quad \forall x > 0$$

Properties of Expectation:

- Linearity: $E(a * X + b) = a * E(X) + b$, where ' a ' and ' b ' are constants. In other words, the expectation of a linear combination of random variables is equal to the linear combination of their individual expectations.
- Constant: $E(c) = c$, where ' c ' is a constant. The expectation of a constant is equal to the constant itself.
- Indicator Function: $E(I_A) = P(A)$, where I_A is the indicator function of an event A . The expectation of an indicator function is equal to the probability of the event.



Variance (Second Central Moment)

The variance (Var) of a random variable X measures how spread out the values of X are around its expectation.

For a discrete random variable X with expectation $E(X)$, the variance is given by:

$$Var(X) = \sum \left[(x - E(X))^2 * P(X = x) \right] \quad \forall x > 0$$

For a continuous random variable X with expectation $E(X)$, the variance is given by:

$$Var(X) = \int \left[(x - E(X))^2 * f(x) \right] dx \quad \forall x > 0$$

Properties of Variance:

- Linearity: $Var(aX + b) = a^2 * Var(X)$, where ' a ' and ' b ' are constants. When a random variable is scaled or shifted, the variance is scaled by the square of the constant.
- Constant: $Var(c) = 0$, where ' c ' is a constant. The variance of a constant is zero since there is no variability.
- Independence: If X and Y are independent random variables, then $Var(X + Y) = Var(X) + Var(Y)$. For independent random variables, the variance of their sum is the sum of their individual variances.



Skewness (Third Central Moment)

Skewness is a statistical measure that quantifies the asymmetry of a probability distribution or a dataset. It helps to understand the extent to which the distribution is shifted to the left or right relative to its center.

A positively skewed distribution (right-skewed) has a long tail to the right, meaning it extends more towards higher values, while a negatively skewed distribution (left-skewed) has a long tail to the left, extending more towards lower values. A symmetric distribution has zero skewness.

Skewness can be calculated using different formulas, but one of the commonly used measures is based on the third standardized moment of the distribution. The formula for skewness (S) using the third standardized moment is as follows:

$$S = \frac{E(x - \mu)^3}{\sigma^3}$$

Where:

σ^3 is the standard deviation of the distribution raised to the power of 3.

Note:

If the skewness value is positive, the distribution is positively skewed (right-skewed).

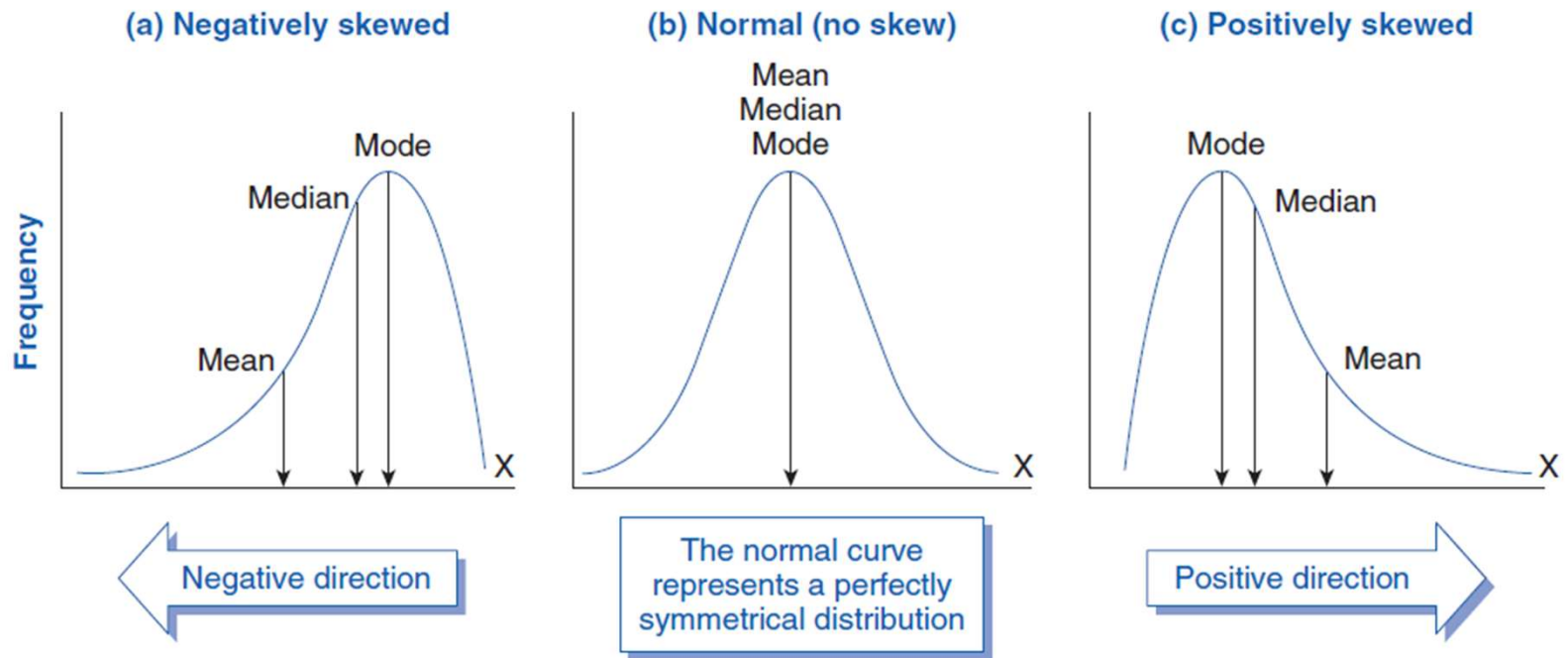
If the skewness value is negative, the distribution is negatively skewed (left-skewed).

If the skewness value is close to zero, the distribution is approximately symmetric.

Skewness and Kurtosis (Third and fourth moments)



Skewness (Third Central Moment)





Kurtosis (Fourth Central Moment)

Kurtosis is a statistical measure that quantifies the degree of peaked-ness or flatness of a probability distribution compared to the normal distribution. It provides information about the tails of the distribution and whether it has more or fewer extreme values than the normal distribution.

A high kurtosis value indicates heavy tails, meaning the distribution has more extreme values than the normal distribution. On the other hand, a low kurtosis value indicates light tails, suggesting the distribution has fewer extreme values.

The formula for kurtosis (K) using the fourth central moment is as follows:

$$K = \frac{E(x - \mu)^4}{\sigma^4}$$

Where:

σ^4 is the variance of the distribution raised to the power of 4.

Note:

For a normal distribution, the kurtosis value is 3 and it is called mesokurtic .

If the distribution has higher kurtosis than 3, it is said to have positive kurtosis (leptokurtic), indicating more peaked-ness and heavier tails.

If the kurtosis is less than 3, the distribution has negative kurtosis (platykurtic), suggesting less peaked-ness and lighter tails.

It's important to note that kurtosis is a dimensionless measure, meaning it does not have any units of measurement. The comparison to the normal distribution's kurtosis of 3 is a relative measure.



Skewness and Kurtosis (Third and fourth moments)

Kurtosis (Fourth Central Moment)

